

PROJECT REPORT OF CO-OP Project at Industry (Module-2)
ON
Towards Anatomy-Agnostic MRI Artifact Detection Using Unsupervised Learning
submitted in partial fulfilment of the requirements for the award of degree of
BACHELOR OF ENGINEERING
In
COMPUTER SCIENCE AND ENGINEERING

Submitted by:

Soumya Makkar

2210992401

Supervised By:

Dr. Gurpreet Singh

Assistant Professor

Chitkara University



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CHITKARA UNIVERSITY INSTITUTE OF ENGINEERING AND TECHNOLOGY
CHITKARA UNIVERSITY, PUNJAB, INDIA

CONTENTS

S.No.	Title	Page No.
	Acknowledgement	4
	Abstract	5
1	Introduction	6
2	Methodology	8
3	Tools and Technologies	13
4	Implementation	15
5	Major Findings/Outcomes/Output/Results	24
6	Conclusion	32
	References	34

DECLARATION

I hereby certify that the work which is being presented in the project report entitled “**Towards Anatomy-Agnostic MRI Artifact Detection Using Unsupervised Learning**” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr. Gurpreet Singh**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura

Soumya Makkar

Date: 17 May 2026

2210992401

This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

Dr. Gurpreet Singh

Assistant Professor

Department of Computer Science and Engineering

Chitkara University Institute of Engineering and Technology,

Chitkara University, Punjab, India

ACKNOWLEDGEMENT

I express my sincere gratitude to all those who have contributed to the successful completion of this research work titled “**Towards Anatomy-Agnostic MRI Artifact Detection Using Unsupervised Learning**”.

First and foremost, I would like to express my deep sense of gratitude and respect to my class mentor, **Dr. Gurpreet Singh**, for his constant guidance, encouragement, and valuable support throughout the course of this work. His motivation and academic mentorship have been instrumental in helping me complete this research successfully.

I would also like to extend my heartfelt thanks to the faculty members of **Chitkara University Institute of Engineering and Technology, Chitkara University, Patiala**, for their invaluable guidance, constructive suggestions, and continuous academic support during this study. Their insights greatly contributed to the improvement and completion of this research.

I am deeply thankful to **Chitkara University** for providing a conducive academic environment, institutional support, and the necessary resources required to carry out this work effectively.

I would further like to acknowledge the creators and contributors of the publicly available datasets used in this study, namely **IXI, FastMRI, MR-ART, and KMAR-50K**. Their efforts in making these valuable medical imaging datasets openly accessible played a significant role in enabling this research.

In addition, I sincerely acknowledge the contributions of researchers and scholars whose prior studies in the fields of MRI artifact detection, medical image quality assessment, anomaly detection, and self-supervised learning provided the theoretical and technical foundation for the present work.

Lastly, I would like to express my heartfelt appreciation to my family and well-wishers for their unwavering support, patience, and encouragement throughout the completion of this research and report.

ABSTRACT

Magnetic Resonance Imaging (MRI) plays a vital role in modern clinical diagnosis due to its non-invasive nature and ability to provide detailed anatomical information. However, MRI scans are often affected by artifacts caused by factors such as patient motion, hardware limitations, and acquisition-related inconsistencies. These artifacts can significantly degrade image quality, reduce diagnostic reliability, and in some cases even mimic pathological findings. Since deep learning systems used in medical imaging are highly sensitive to image quality, the automated detection of MRI artifacts has become an important research problem.

Most existing MRI artifact detection approaches rely heavily on supervised learning, which requires large amounts of labelled data. In medical imaging, such annotations are expensive, time-consuming, and often subjective. Moreover, many prior methods are designed for a single anatomy or imaging sequence, limiting their ability to generalize across diverse clinical settings.

This research presents a systematic comparative study of supervised, unsupervised, and self-supervised learning paradigms for MRI artifact detection under a unified experimental framework. The study evaluates supervised baselines such as ResNet-18, unsupervised anomaly detection models such as autoencoders and PatchCore, and self-supervised representation learning approaches including SimCLR, DINO, and Masked Autoencoders. Experiments were conducted using multiple publicly available MRI datasets spanning both brain and knee anatomies to examine cross-dataset and cross-anatomy generalization.

The findings indicate that most methods struggle to achieve strong performance on real MRI motion artifacts under domain shift conditions. Among the evaluated approaches, PatchCore with k-nearest neighbour scoring produced the most consistent above-chance performance on real artifact datasets. The study also reveals important challenges such as shortcut learning in supervised models, representational collapse in self-distillation methods, and a clear performance gap between synthetic and real artifacts. Overall, the work highlights both the promise and the current limitations of label-free approaches for anatomy-agnostic MRI artifact detection, and emphasizes the need for more robust, generalizable, and clinically realistic evaluation strategies.

1. Introduction

Magnetic Resonance Imaging (MRI) has become one of the most important imaging modalities in modern clinical practice due to its non-invasive nature and its ability to provide highly detailed anatomical and soft-tissue information without the use of ionizing radiation. Its wide applicability across neurological, musculoskeletal, abdominal, and oncological imaging has made it indispensable for diagnosis, treatment planning, and disease monitoring. However, despite its clinical value, MRI remains highly sensitive to various sources of image degradation that arise during the acquisition process. These degradations, commonly referred to as artifacts, can substantially reduce image quality and compromise the diagnostic usefulness of a scan.

MRI artifacts may arise from several physical, physiological, and acquisition-related factors such as patient motion, hardware instability, magnetic field inhomogeneity, aliasing, ghosting, and bias field effects. Unlike random image noise, many MRI artifacts exhibit structured patterns that can obscure anatomical detail, mimic pathology, or make scans difficult to interpret reliably. In severe cases, artifacts may render a scan clinically unusable, requiring repeat acquisitions and thereby increasing scanning time, operational cost, and patient discomfort. Motion artifacts, in particular, remain one of the most persistent challenges in clinical MRI, especially in emergency, pediatric, elderly, or uncooperative patient populations. Their impact extends beyond radiological interpretation and affects downstream computational tasks as well.

In recent years, the rapid growth of deep learning in medical imaging has enabled substantial progress in tasks such as classification, segmentation, registration, reconstruction, and abnormality detection. However, the success of these methods is highly dependent on the quality of input images. Deep neural networks are often sensitive to subtle distortions in image appearance and may produce unreliable predictions when trained or evaluated on corrupted scans. As a result, automated MRI artifact detection has emerged as an important research problem, not only for maintaining diagnostic quality but also for ensuring the robustness of AI-based medical imaging systems. An effective artifact detection framework can serve as an essential quality control mechanism by identifying problematic scans before they are used for clinical interpretation or downstream algorithmic analysis.

Traditionally, MRI quality control has largely been approached through Image Quality Assessment (IQA) methods, where a scan is assigned an overall quality score. While useful for providing a broad measure of image usability, such approaches often fail to explicitly identify the specific artifacts responsible for degradation. Furthermore, many existing IQA methods are evaluated on synthetically generated corruptions such as blur or added noise, which may not accurately reflect the complex and structured

nature of real-world MRI artifacts. This creates an important gap between benchmark performance and practical clinical utility.

To address this challenge, more recent research has increasingly focused on direct artifact detection using machine learning and deep learning techniques. Supervised methods have shown promising performance in identifying artifacts and grading their severity, but they depend heavily on large, expertly annotated datasets. In medical imaging, such labelled datasets are often difficult and expensive to obtain, and artifact annotations may also be subjective. In addition, many existing studies are restricted to a single anatomy, imaging sequence, or acquisition setting, which limits their ability to generalize across diverse clinical conditions. A model trained only on one anatomical region may inadvertently learn anatomy-specific features instead of learning to recognize artifact-related distortions in a more generalizable manner.

These limitations highlight the need for more scalable and anatomy-agnostic approaches to MRI artifact detection. In this context, unsupervised and self-supervised learning offer a promising alternative. Instead of relying on large labelled artifact datasets, these paradigms learn meaningful representations from clean or unlabelled images and frame artifact detection as an anomaly detection problem. Such an approach is particularly attractive in medical imaging, where clean scans are more readily available than consistently annotated artifact data. Moreover, label-free learning strategies have the potential to improve generalization across anatomies, scanners, and acquisition protocols by focusing on structural consistency rather than dataset-specific labels.

The present study is motivated by this broader need for label-efficient and generalizable MRI quality control systems. It presents a systematic evaluation of supervised, unsupervised, and self-supervised learning paradigms for artifact detection under a unified experimental framework. By incorporating multiple open datasets spanning both brain and knee MRI, the work moves beyond single-anatomy evaluation and explicitly investigates cross-dataset and cross-anatomy generalization. The study compares reconstruction-based methods, feature-based anomaly detection approaches, and representation learning techniques to understand their relative strengths and limitations when applied to real-world artifact detection scenarios.

Thus, this research aims not only to assess model performance but also to examine a deeper question: whether it is possible to learn anatomy-independent representations of MRI quality without relying on artifact-specific annotations. In doing so, the study contributes to the ongoing effort to develop practical solutions for automated MRI quality assessment that are better aligned with real-world clinical deployment.

2. Methodology

The methodology of this study was designed to systematically evaluate the feasibility of anatomy-agnostic MRI artifact detection using supervised, unsupervised, and self-supervised learning paradigms under a unified framework. Since the central objective of the research was not only to compare model performance but also to examine generalization across anatomies, scanners, and datasets, the methodological pipeline was built around five major components: dataset selection, preprocessing and harmonization, test set construction, model selection, and experimental design. The overall approach reformulates MRI artifact detection as both a binary classification problem and an anomaly detection problem, depending on the learning paradigm under consideration.

2.1 Dataset Selection

To ensure diversity in anatomy, acquisition settings, and scanner characteristics, a multi-cohort dataset was curated using publicly available MRI datasets. The datasets were selected to allow training predominantly on clean MRI scans and evaluation on artifact-corrupted scans, thereby enabling both supervised and label-free learning experiments.

The study focused on two anatomies: brain MRI and knee MRI, with data drawn from four primary datasets (see Table 1).

2.1.1 Rationale for Dataset Selection

The selected datasets collectively enabled:

- training on clean data only for unsupervised and self-supervised methods,
- evaluation on real artifact distributions,
- analysis across brain and knee anatomies,
- and study of cross-scanner and cross-dataset generalization.

Thus, the dataset design directly supported the study’s main goal of exploring anatomy-independent MRI artifact detection.

Table 1: Summary of datasets used in this study, highlighting variations in anatomy, acquisition settings,

Dataset	Anatomy	Scans/Volumes	(Clean/Artifact)	Sequences	Field Strength	Manufacturer(s)
IXI	Brain	~600	Clean	T1, T2	1.5T, 3T	Philips, GE
FastMRI	Knee	200	Clean	PD, PD-FS, T2-FS	1.5T, 3T	Siemens
MR-ART	Brain	~445	Artifact	T1	3T	Siemens
KMAR-50K	Knee	1,444 (paired)	Artifact	PD, T1, T2 (TSE/FSE)	1.5T, 3T	Siemens, Alltech

and availability of artifact annotations

2.2 Data Preprocessing and Harmonization

Because the selected datasets differed substantially in file format, spatial resolution, dimensionality, and anatomy, a unified preprocessing pipeline was designed to convert all images into a common representation suitable for model training and testing.

A major design decision in this study was to avoid anatomy-specific preprocessing operations such as skull stripping, registration, or segmentation, as these steps would undermine the anatomy-agnostic nature of the proposed framework. Instead, a lightweight but standardized pipeline was used.

2.2.1 Preprocessing of FastMRI Knee Data

The FastMRI knee dataset was stored in HDF5 format as volumetric scans. The preprocessing steps were as follows:

- a. Volumes were extracted from the .hdf5 files as 3D arrays.
- b. Each volume was center-cropped to a size of $192 \times 192 \times 32$.
- c. Percentile-based intensity normalization was applied on a per-volume basis:
 - i. intensities were clipped to the range $[p1, p99]$
 - ii. then scaled to $[0, 1]$
- d. The volumes were decomposed into axial 2D slices.
- e. Slices with a mean intensity below 0.01 were discarded to remove nearly empty or non-informative slices.

This process ensured that all FastMRI slices had consistent dimensions and normalized intensity values.

2.2.2 Preprocessing of IXI Brain Data

The IXI brain MRI volumes required a different initial preprocessing stage due to their spatial characteristics. The following steps were applied:

- a. Volumes were resampled to 1 mm isotropic resolution.
- b. Each volume was center-cropped to $192 \times 192 \times 128$.
- c. Balanced sampling was performed across T1 and T2 modalities.
- d. Like FastMRI, percentile normalization was applied:
 - i. clipping intensities to $[p1, p99]$
 - ii. scaling them to $[0, 1]$

- e. Volumes were then decomposed into axial slices.
- f. Slices with a mean intensity less than 0.01 were removed.
- g. A stratified slice sampling strategy was introduced:
 - i. 70% of selected slices came from the central regions of the volume
 - ii. 30% came from peripheral regions

This additional sampling step ensured that the brain data retained both anatomically informative central slices and diverse peripheral slices.

2.2.3 Preprocessing of Artifact Datasets

The artifact datasets required preprocessing according to whether they were already in 2D slice form or in 3D volumetric form.

MR-ART and MRI-BT

For datasets already stored as individual slices:

- Images were loaded as grayscale float32 arrays
- center cropped to 192×192
- No geometric distortion was introduced during resizing

KMAR-50K

For 3D volumetric artifact data:

- Volumes were loaded from NIfTI files
- Center cropped to $192 \times 192 \times 32$
- Intensity normalization and slice filtering steps, like those of the clean datasets, were applied
- Axial slices were extracted for further use

2.2.4 Final Harmonized Representation

After preprocessing, every sample in the study—regardless of its source—was represented as a 2D grayscale MRI slice with:

- a. spatial size of 192×192
- b. intensity normalized to $[0,1]$
- c. binary label assigned as:
 - i. 0 for clean slices
 - ii. 1 for artifact-corrupted slices

This harmonized representation made it possible to train and evaluate all models under a common experimental setting.

2.5 Test Set Construction and Dataset Balancing

A rigorous test set construction strategy was followed to ensure fairness, reproducibility, and strict separation between training and evaluation data.

2.5.1 Training and Test Partitioning

The clean datasets (IXI and FastMRI) were used:

- for self-supervised and unsupervised training
- and as negative samples in supervised training

The artifact datasets (MR-ART and KMAR) were split into:

- One subset used in supervised training
- another completely disjoint subset reserved only for testing

This ensured that no artifact sample encountered during training appeared again during evaluation.

2.5.2 Balancing of Artifact and Clean Samples

To avoid evaluation bias caused by unequal dataset sizes, the MR-ART and KMAR datasets were balanced by equalizing the number of clean and artifact slices across datasets. The artifact sample count was selected using the minimum available count between datasets. A similar balancing strategy was applied for clean samples. Subsampling was performed with a fixed random seed, ensuring consistent data selection across repeated runs.

2.5.3 Use of Simulated Artifact Dataset

The MRI-BT simulated artifact dataset was not used for supervised training. Instead, it was used only during the evaluation of the self-supervised and anomaly detection models. This enabled the study to compare performance on real artifacts versus synthetically generated degradations.

2.6 Model Selection

The research aimed to compare three major learning paradigms for MRI artifact detection:

- a. Supervised learning
- b. Unsupervised anomaly detection
- c. Self-supervised representation learning

The selected models were chosen to represent diverse methodological principles and to test how different forms of representation learning behave under domain shift and anatomy variation.

2.6.1 Supervised Learning Models

- a. **ResNet-18:** The primary supervised baseline was **ResNet-18**, a convolutional neural network known for its residual skip connections, which help overcome the vanishing gradient problem and allow deeper training stability. ResNet-based architectures are widely used in medical imaging because of their strong optimization behaviour and robust visual feature extraction.

2.6.2 Unsupervised Anomaly Detection Models

- a. **Autoencoders / Denoising Autoencoders:** Autoencoders were used as a classical anomaly detection baseline. They consist of:
 - i. an encoder, which compresses the image into a latent representation
 - ii. a decoder, which reconstructs the input

The underlying assumption is that a model trained only on clean images will reconstruct normal images well but produce higher reconstruction errors for anomalous or artifact-corrupted inputs.

A **denoising autoencoder (DAE)** variant was also explored to improve robustness.

- b. **PatchCore:** PatchCore was selected as a feature-based anomaly detection method. Instead of reconstructing images, PatchCore builds a memory bank of representative patch-level features extracted from clean images using a pretrained backbone. During inference, a test slice is compared against this memory bank, and anomaly scores are derived from nearest-neighbour distances. This method is especially useful for identifying localized deviations and has shown strong performance in industrial anomaly detection tasks.

2.6.3 Self-Supervised Learning Models

- a. **SimCLR:** SimCLR is a contrastive learning framework that learns representations by maximizing agreement between two different augmented views of the same image while separating them from other samples in the batch. It was selected because of its ability to learn robust visual representations without labels.
- b. **DINO:** DINO is a self-distillation-based framework using a student-teacher structure without negative samples. It was included because of its ability to learn globally meaningful and semantically structured representations, particularly when implemented with vision transformers.
- c. **Masked Autoencoders (MAE):** MAE is a self-supervised transformer-based model that learns to reconstruct randomly masked portions of the input image. Unlike standard autoencoders, it focuses more on representation learning than exact pixel reconstruction, making it relevant for both reconstruction-based and feature-based anomaly detection.

3. Tools and Technologies

The implementation of this study was carried out using a Python-based deep learning and medical image analysis stack. Python was used as the primary programming language because it provides strong support for machine learning development, image preprocessing, numerical computation, and reproducible experimentation.

For deep learning model development, PyTorch was used as the main framework. It was employed to implement and train the supervised ResNet-18 baseline, the SimCLR self-supervised framework, the denoising autoencoder, and the reconstruction and feature-extraction pipelines required for anomaly detection. PyTorch was selected because of its flexibility in defining custom training loops, loss functions, augmentation pipelines, and evaluation procedures across multiple experimental settings.

Torchvision was used alongside PyTorch for model backbones, image transformations, and augmentation utilities. It supported the use of convolutional architectures such as ResNet-based encoders and helped maintain a consistent preprocessing and transformation pipeline across methods.

For transformer-based self-supervised learning methods such as DINO and Masked Autoencoders (MAE), Vision Transformer implementations compatible with the PyTorch ecosystem were used.

These were required for patch-based image tokenization, masked reconstruction, student-teacher self-distillation, and embedding extraction from ViT-Small/16 backbones.

For data handling and preprocessing, NumPy was used for numerical array manipulation and intensity normalization, while OpenCV was used for image resizing, center cropping, and slice-level geometric processing. The h5py library was used to load fastMRI data stored in HDF5 format, and nibabel was used to read volumetric MRI data stored in NIfTI format for datasets such as IXI, MR-ART, and KMAR. These tools enabled all datasets to be converted into a common 2D slice-based representation of size 192×192 .

For data organization, metadata handling, and results tabulation, pandas was used extensively. It supported the creation of manifest files, train-test splits, experiment logs, and structured performance summaries across all methods, training variants, and test datasets.

For evaluation and classical machine learning utilities, scikit-learn was used. It was employed to compute performance metrics including AUROC, AUPRC, F1-score, precision, and recall. It was also used for nearest-neighbor based anomaly scoring in feature space, particularly in kNN-based evaluation for SimCLR, DINO, MAE, and PatchCore-style scoring pipelines.

For scientific visualization and qualitative analysis, Matplotlib and Seaborn were used. These libraries were used to generate precision-recall curves, AUROC comparison plots, confusion matrices, feature distribution plots, and other comparative visualizations used in the analysis of results.

The experiments were executed in cloud-based notebook environments, primarily Kaggle, which provided access to Tesla T4 GPUs for training most models. According to the experimental setup, PatchCore evaluation additionally made use of TPU v3 resources for efficient feature extraction and large-scale memory-bank inference. These platforms made it possible to train and evaluate all 18 model configurations under limited local hardware availability.

To ensure reproducibility, all experiments were conducted with a fixed random seed of 42. The same seed was maintained across preprocessing, dataset subsampling, training, and evaluation so that comparisons between supervised, unsupervised, and self-supervised methods remained fair and consistent.

If you want, I can also convert this into a more compact report version with subheadings like: Programming Language, Frameworks, Libraries, Hardware Platform, and Reproducibility.

4. Implementation

The implementation of the proposed research work was carried out as a structured experimental pipeline for evaluating supervised, unsupervised, and self-supervised approaches for MRI artifact detection under a unified setup. The overall implementation was designed to maintain fairness across methods, ensure reproducibility, and allow direct comparison between different learning paradigms across multiple anatomical domains. Since the central aim of the work was to investigate anatomy-agnostic artifact detection, the implementation emphasized common preprocessing, balanced dataset construction, standardized training settings, and consistent evaluation protocols.

4.1 Implementation Environment

The complete experimental framework was implemented using deep learning-based training and evaluation pipelines suitable for handling large-scale MRI slice data. The experiments involved both convolutional neural networks and transformer-based models, depending on the learning paradigm being evaluated.

Training and evaluation were conducted using the following computing resources:

- a Tesla T4 GPU (15.6 GB memory) for:
 - i supervised ResNet-based classification,
 - ii SimCLR,
 - iii Masked Autoencoder (MAE),
 - iv Denoising Autoencoder (DAE)
- b TPU v3 for:
 - i PatchCore-based anomaly detection experiments

The use of dedicated hardware resources ensured efficient execution of the complete model suite while maintaining uniformity in training settings as much as possible across experiments.

To ensure reproducibility, a fixed random seed of **42** was used throughout:

- a. dataset subsampling,
- b. slice selection,
- c. train-validation partitioning,
- d. and model evaluation.

4.2 Data Handling and Input Representation

The implementation was built on a harmonized slice-based MRI representation. Since the source datasets varied in anatomy, dimensionality, acquisition format, and scanner configuration, all images were converted into a unified format before being passed to the models.

After preprocessing, each MRI sample was represented as:

- a a 2D grayscale slice
- b spatial resolution of 192×192
- c intensity normalized to the range $[0,1]$

Each slice was then assigned a binary label where:

- a 0 indicated a clean image
- b 1 indicated an artifact-corrupted image

This common representation allowed all models to operate under a unified experimental setting, regardless of the original dataset source.

4.3 Implementation Setup

The training was conducted under three anatomical variants:

1. Knee-only
2. Brain-only
3. Combined multi-anatomy

4.4 Supervised Model Implementation

In the supervised branch, artifact detection was implemented as a binary slice classification task.

Architecture

The primary supervised classifier was based on **ResNet-18**, adapted for single-channel MRI input. Since MRI slices are grayscale rather than RGB, the input layer was modified accordingly. The model produced a scalar logit for each input slice, which was converted into artifact probability through a sigmoid activation.

Although EfficientNet-B0 was also part of the broader methodological consideration, the primary supervised implementation described in the experiments centered on ResNet-18.

Loss Function

To account for class imbalance between clean and artifact slices, a **weighted binary cross-entropy loss** was implemented. The positive class weighting was defined as:

$$w = \frac{N_{\text{neg}}}{N_{\text{pos}}}$$

In the reported supervised setup:

- Artifact positives were sampled as **7.5% of the clean training count**
- The positive ratio was **5.65%**
- **pos_weight = 16.68**

This weighting helped prevent the classifier from being biased toward the dominant clean class.

Optimization Settings

The supervised model was trained using:

- **AdamW optimizer**
- initial learning rate: **3×10^{-4}**
- weight decay: **1×10^{-4}**
- total epochs: **20**
- mini-batch size: **64**
- gradient accumulation: **4 steps**
- effective batch size: **256**
- gradient clipping: **1.0**
- mixed precision training

A **cosine annealing learning rate schedule** with **2 epochs of linear warmup** was used.

Training Variants

The supervised classifier was trained under:

- a knee-only configuration
- b brain-only configuration
- c combined multi-anatomy configuration

Manifest-based training and validation splits were used to maintain experiment consistency.

4.5 SimCLR Implementation

The SimCLR framework was implemented as a self-supervised contrastive representation learning pipeline for learning anatomy-invariant MRI features without labels.

Encoder and Projection Head

The SimCLR model used:

- a a ResNet-18 encoder adapted for single-channel input
- b a two-layer MLP projection head

For each input slice, two independent augmented views were generated and passed through the shared encoder and projection head.

Augmentation Strategy

A domain-specific augmentation pipeline was implemented, including:

- a. affine transformations
- b. elastic deformations
- c. intensity perturbations
- d. Gaussian noise
- e. bias field simulation
- f. random cutout

These augmentations were chosen to encourage the encoder to learn robust structural features while becoming less sensitive to superficial intensity or spatial variations.

Loss and Training

The embeddings were L2-normalized and optimized using the **NT-Xent contrastive loss** with temperature:

$$\tau = 0.5$$

Training settings were:

- a. optimizer: **AdamW**
- b. learning rate: 1×10^{-3}
- c. weight decay: 1×10^{-4}
- d. epochs: **20**
- e. batch size: **64**

- f. gradient accumulation factor: **4**
- g. effective batch size: **256**

After training, the projection head was discarded and the encoder embeddings were used for anomaly detection through k-nearest neighbor (kNN) scoring.

4.6 Denoising Autoencoder Implementation

The denoising autoencoder (DAE) was implemented as a reconstruction-based unsupervised anomaly detection baseline.

Model Structure

The DAE followed an encoder-decoder architecture in which:

- a. The encoder compressed the input MRI slice into a latent representation
- b. The decoder reconstructed the image from this latent code

The encoder bottleneck used a compressed representation derived from convolutional feature blocks.

Working Principle

The implementation was based on the idea that a model trained primarily on clean MRI data should reconstruct normal slices more effectively than artifact-corrupted slices. During testing:

- a. The reconstruction error was used as one anomaly signal
- b. Latent embeddings were also used for **kNN-based anomaly scoring**

Thus, two evaluation modes were considered: DAE (Recon) & DAE (kNN).

Training Configuration

The DAE was trained using:

- a. **AdamW optimizer**
- b. weight decay: 1×10^{-4}
- c. cosine learning rate schedule with warmup
- d. **20** training epochs
- e. effective batch size of **256**

4.6 PatchCore Implementation

PatchCore was implemented as a feature-based anomaly detection framework operating on a memory bank of normal patch embeddings.

Feature Extraction

PatchCore used features extracted from a pretrained convolutional backbone: an ImageNet-based feature extractor. Intermediate feature maps were collected and locally aggregated to form patch-level embeddings.

Memory Bank Construction

A memory bank of representative clean patch features was created from the training data. To improve computational efficiency and reduce redundancy coreset subsampling was applied. This ensured that the memory bank retained only representative samples of the clean feature distribution.

Inference Procedure

At test time:

- a. patch features were extracted from the input slice
- b. Each feature was compared with the memory bank
- c. nearest-neighbor distances were computed
- d. Anomaly scores were aggregated at the image level

Two forms of PatchCore scoring were evaluated:

- a. PatchCore (Patch)
- b. PatchCore (kNN)

The kNN-based global scoring variant showed the strongest real-artifact performance in the reported experiments.

4.7 Masked Autoencoder (MAE) Implementation

The MAE model was implemented as a transformer-based self-supervised learning framework.

Architecture

The implementation used a:

- **ViT-Small/16 encoder**
- embedding dimension: **384**
- transformer depth: **12 layers**

Each input image was divided into non-overlapping patches.

Masking Strategy

A random **75% masking ratio** was used, meaning that only 25% of patches were visible to the encoder during training. The decoder reconstructed the masked content.

Training Objective

The model minimized reconstruction error only over masked patches. After training:

- The decoder was discarded
- The encoder was used for feature extraction

Training Settings

MAE was trained for:

YYY20 epochs

- AdamW optimizer**
- effective batch size: **256**
- cosine learning rate schedule with warmup
- weight decay: **5×10^{-2}**

Three training variants were implemented:

- knee-only
- brain-only
- combined

4.8 DINO Implementation

DINO was implemented as a **self-distillation-based self-supervised learning model** using a student-teacher transformer architecture.

Backbone

Both student and teacher used:

- a. **ViT-Small/16**
- b. embedding dimension: **384**
- c. 12 transformer layers

The teacher network parameters were updated as an exponential moving average of the student parameters.

Multi-Crop Augmentation

The implementation used the multi-crop strategy:

- a. **2 global views** of size **192×192**
- b. **6 local views** of size **96×96**

The teacher processed only global views, while the student processed all views.

Collapse Prevention

To reduce the risk of representational collapse, the implementation included:

- a. centering
- b. temperature scaling
- c. progressive teacher momentum updates

After training, the teacher encoder was used to extract L2-normalized features, and anomaly detection was performed using **kNN scoring**.

Representation Health Monitoring

An important part of the DINO implementation was post-training representation analysis using:

- a. feature standard deviation
- b. mean pairwise cosine similarity
- c. effective rank from singular value decomposition
- d. t-SNE visualization
- e. nearest-neighbor retrieval

This analysis was important because the brain-only DINO variant showed clear feature collapse.

4.8 Common Training Strategy Across Models

To make the comparison fair, most models shared a common implementation protocol.

4.8.1 Shared Hyperparameters

Across methods, the following were largely standardized:

- a. **20 epochs**
- b. effective batch size: **256**
- c. **AdamW optimizer**
- d. cosine learning rate schedule
- e. **2-epoch warmup**
- f. fixed seed: **42**

10.2 Training Time

The total implementation effort involved training:

- a. **6 methods**
- b. across **3 anatomical configurations**
- c. giving **18 model instances**

The total training time was approximately:

- a. **24 hours** across all completed runs

The total evaluation time was:

- a. **133 minutes** across **54 evaluation runs**

4.9 Evaluation Implementation

The evaluation pipeline was implemented consistently across all methods and test sets.

4.9.1 Metrics

The following metrics were computed:

- a. AUROC
- b. AUPRC
- c. F1-score
- d. Precision
- e. Recall
- f. Detection Rate
- g. Mean anomaly score

For MR-ART and KMAR, thresholds were selected using the **Youden-optimal threshold**.

For the simulated artifact-only dataset, AUROC was not defined, and the detection rate was reported using the threshold obtained from the MR-ART evaluation

Comparative Evaluation Design

Each trained model was evaluated on:

- a. MR-ART
- b. KMAR
- c. MRI-BT simulated artifacts

This enabled analysis of:

- a. within-anatomy generalization
- b. cross-anatomy generalization
- c. cross-dataset robustness
- d. synthetic vs real artifact sensitivity
- e. representational collapse in SSL method

5. Major Findings/Outcomes/Output/Results

The quantitative evaluation of the proposed framework was conducted across multiple models, training configurations, and test datasets in order to systematically assess the effectiveness of supervised, unsupervised, and self-supervised approaches for MRI artifact detection. Performance was measured primarily using AUROC, AUPRC, and F1-score on the real-artifact datasets MR-ART and KMAR, while detection rate was reported for the simulated artifact dataset, where AUROC was not applicable.

5.1 Quantitative Results

Table 2: Quantitative Results Across All Models and Test Sets

Method	Training Variant	MR-ART AUROC	KMAR AUROC	Simulated Detection Rate
Supervised	Knee	0.4998	0.5026	0.8375

Method	Training Variant	MR-ART AUROC	KMAR AUROC	Simulated Detection Rate
Supervised	Brain	0.5054	0.4978	0.1199
Supervised	Combined	0.5010	0.5029	0.0915
SimCLR (kNN)	Knee	0.4813	0.4783	0.0268
SimCLR (kNN)	Brain	0.5691	0.5020	0.1720
SimCLR (kNN)	Combined	0.5361	0.4870	0.8048
MAE (kNN)	Knee	0.4165	0.5088	0.0059
MAE (kNN)	Brain	0.2931	0.4801	0.2182
MAE (kNN)	Combined	0.4544	0.5117	0.0003
MAE (Recon)	Knee	0.4593	0.4851	0.3476
MAE (Recon)	Brain	0.4672	0.4758	0.9842
MAE (Recon)	Combined	0.4594	0.4759	0.9885
DINO (kNN)	Knee	0.3549	0.4909	0.5173
DINO (kNN)	Brain	0.5674	0.4913	0.7777
DINO (kNN)	Combined	0.5677	0.4988	0.9665
DAE (kNN)	Knee	0.3270	0.4994	0.0818
DAE (kNN)	Brain	0.4443	0.4941	0.2412
DAE (kNN)	Combined	0.4357	0.5003	0.2990
DAE (Recon)	Knee	0.4359	0.4833	0.6307
DAE (Recon)	Brain	0.4971	0.4846	0.5681
DAE (Recon)	Combined	0.5167	0.4942	0.5484
PatchCore (Patch)	Knee	0.2902	0.4507	0.2487
PatchCore (Patch)	Brain	0.5149	0.4729	0.9346
PatchCore (Patch)	Combined	0.4069	0.4615	0.3039
PatchCore (kNN)	Knee	0.4068	0.5269	0.0051
PatchCore (kNN)	Brain	0.6730	0.5261	0.9226
PatchCore (kNN)	Combined	0.6719	0.5283	0.9205

Table 3. Cross-anatomy Generalization Performance

Method	Brain → KMAR AUROC	Knee → MR-ART AUROC
Supervised	0.4978	0.4998
SimCLR (kNN)	0.5020	0.4813
MAE (kNN)	0.4801	0.4165
MAE (Recon)	0.4758	0.4593
DINO (kNN)	0.4913	0.3549
DAE (kNN)	0.4941	0.3270
DAE (Recon)	0.4846	0.4359
PatchCore (Patch)	0.4729	0.2902
PatchCore (kNN)	0.5261	0.4068

5.2 Major Findings

The present study conducted a systematic evaluation of supervised, unsupervised, and self-supervised learning methods for MRI artifact detection across multiple datasets, anatomical regions, and training configurations. The results provide important insights into the practical challenges of anatomy-agnostic MRI quality control and reveal that real-world artifact detection remains significantly more difficult than expected. The major findings of the work are presented below.

5.2.1. Overall Performance on Real Artifact Datasets

One of the most significant outcomes of the study was that most evaluated methods failed to achieve strong discriminative performance on real MRI artifact datasets. Across MR-ART and KMAR, the majority of AUROC values remained close to **0.5**, indicating performance near random chance. This suggests that detecting real MRI motion artifacts in a generalized and anatomy-independent way is a highly challenging task.

Although the study included a diverse range of models—supervised classifiers, reconstruction-based anomaly detectors, and feature-based self-supervised methods—none of them demonstrated consistently high performance across both real test datasets. This result is important because it shows that current learning-based methods, especially when trained under cross-dataset and cross-anatomy conditions, are still not sufficiently robust for reliable clinical deployment.

5.2.2. PatchCore with kNN Scoring Achieved the Best Real-Artifact Performance

Among all evaluated methods, PatchCore with global k-nearest-neighbor scoring emerged as the most effective approach for real MRI artifact detection. It achieved the strongest above-chance results, particularly on the MR-ART dataset, where the **brain-trained PatchCore-kNN model reached an AUROC of approximately 0.673**, while performance on KMAR remained around **0.53**.

This finding suggests that **feature-memory-based anomaly detection** is comparatively better suited than reconstruction-only or purely supervised methods in this setting. Since PatchCore relies on a pretrained feature extractor and a memory bank of normal features rather than directly optimizing an end-to-end classifier, it appears to be less vulnerable to overfitting and domain-specific shortcut learning. However, even this best-performing method did not achieve sufficiently high performance for dependable real-world artifact screening, which highlights the difficulty of the task.

5.2.3. Supervised Learning Showed Severe Shortcut Learning

A highly important result of the study was that the supervised ResNet-18 baseline performed extremely well during validation but failed on real test data. The model **achieved** near-perfect validation AUROC values close to 1.000, yet its test AUROC on MR-ART and KMAR dropped to approximately **0.499–0.505**, which is effectively chance-level performance.

This outcome clearly indicates the presence of shortcut learning. Instead of truly learning artifact characteristics, the model learned to distinguish the source dataset identity of clean and artifact images. Since clean images came from IXI and fastMRI, while artifact-positive images came from MR-ART and KMAR, the supervised model exploited scanner and dataset-specific visual cues rather than artifact content itself.

This is one of the strongest findings of the study because it demonstrates that supervised MRI artifact detection can be misleading if clean and corrupted samples come from different acquisition domains. Therefore, future supervised studies must use matched clean-artifact pairs from the same scanner, protocol, or subject distribution to avoid this confounding effect.

5.2.4. Real Artifact Detection Was Much Harder Than Simulated Artifact Detection

Another major finding was the strong gap between performance on simulated artifacts and performance on real motion artifacts. Several methods achieved very high detection rates on the simulated MRI-BT artifact dataset yet performed poorly on MR-ART and KMAR.

For example:

- a. **MAE (Reconstruction)** achieved almost **98–99% detection** on simulated artifacts
- b. **DINO (kNN)** and **PatchCore (kNN)** also showed very high simulated artifact detection
- c. Yet these same models produced only modest or near-chance results on real datasets

This finding reveals that synthetic corruptions such as blur, noise, and digital motion smearing are not adequate proxies for real MRI artifacts. Real motion artifacts arise from k-space acquisition errors, producing structured distortions such as ghosting and ringing, while synthetic artifacts often create obvious pixel-level degradation that anomaly detectors can easily identify.

Therefore, the study strongly suggests that evaluation on synthetic artifacts alone can lead to overly optimistic conclusions and may not reflect actual clinical performance.

5.2.5. Within-Anatomy Generalization Was Better Than Cross-Anatomy Generalization

The results also showed that models generally performed better when evaluated on the same anatomy type used during training. This was observed in the within-anatomy setting, such as:

- a. brain-trained models tested on **MR-ART**
- b. knee-trained models tested on **KMAR**

Even in this more favorable setup, performance was still limited, but it was generally stronger than in cross-anatomy evaluation. When models were trained on one anatomy and tested on another, performance often degraded further.

This indicates that, despite the goal of anatomy-agnostic learning, many models still retained anatomy-dependent representations. In other words, they were not fully able to separate artifact-related patterns from normal anatomical structure. This outcome directly supports the central hypothesis of the study that achieving true anatomy-independent artifact detection is difficult and remains an open challenge.

5.2.6. Reconstruction-Based Scoring Was More Anatomy-Agnostic, but Still Weak on Real Data

An important observation from the experiments was that reconstruction-based approaches, especially MAE-Recon and DAE-Recon, showed relatively smaller drops under cross-anatomy evaluation compared to feature-space kNN methods. This suggests that reconstruction error may be more anatomy-agnostic than nearest-neighbor distance in learned feature space.

The likely reason is that reconstruction loss measures how well a model can reproduce image structure at the pixel or patch level, regardless of whether the anatomy is brain or knee. In contrast, feature-space methods depend heavily on how well the training feature manifold matches the target test anatomy and acquisition domain.

However, despite this relative advantage in generalization, the absolute performance of reconstruction-based methods on real artifact datasets still remained weak. Thus, reconstruction-based scoring appears promising conceptually, but in its current form, it is not sufficient for robust real-world artifact detection.

5.2.7. Multi-Anatomy Training Had Mixed Effects Across Models

The study further examined whether training on a combined multi-anatomy dataset improves generalization. The results showed that the effect of multi-anatomy training was not uniform across methods.

Positive effects

For some self-supervised methods, especially **DINO**, combined training significantly improved representation quality and reduced collapse. It also improved simulated artifact detection and preserved reasonable MR-ART performance.

Limited or inconsistent effects

For other methods such as **MAE** and **DAE**, combined training did not lead to a strong or consistent improvement on real artifact datasets. In some cases, the gains were marginal, while in others the method remained near chance.

This outcome indicates that anatomical diversity alone is not enough to guarantee robust artifact detection, although it can act as a useful regularizer for some representation learning frameworks.

5.2.8. Feature Collapse Was Observed in Some Self-Supervised Models

A particularly important technical outcome of the study was the discovery of representational collapse in certain self-supervised settings.

The most severe case was observed in DINO trained only on brain MRI, where feature statistics indicated collapse:

- a. extremely low feature standard deviation
- b. very high pairwise cosine similarity
- c. very low effective rank

This means that the encoder was mapping most images to nearly identical representations, making anomaly detection ineffective. A similar but weaker collapse effect was also observed in the brain-only DAE variant.

In contrast:

- a. **SimCLR** remained stable across all training variants
- b. **MAE** maintained comparatively healthy feature diversity
- c. **PatchCore** did not suffer from collapse because its backbone was frozen

This finding is valuable because it shows that self-supervised learning in medical imaging can be unstable when training data lacks sufficient diversity, and that collapse diagnostics should be treated as an essential part of model evaluation.

5.2.9. SimCLR Produced Stable Representations but Limited Real-World Gains

SimCLR demonstrated one of the healthiest representational behaviors among all tested methods. It avoided collapse across knee-only, brain-only, and combined settings, and its contrastive objective produced stable embeddings.

However, this representational stability did not translate into strong real-artifact detection performance. While SimCLR showed moderate improvements in some settings and performed well on simulated artifacts under combined training, its real-dataset AUROC values still remained modest.

This indicates that learning stable representations is not by itself sufficient; the learned features must also align specifically with clinically meaningful artifact characteristics.

10. The Study Exposed a Strong Train-Test Distribution Gap

A recurring pattern in the results was the presence of a substantial distribution gap between clean training data and clean test data. Clean training samples came from IXI and fastMRI, whereas clean test samples in MR-ART and KMAR were acquired under different scanners, protocols, and acquisition settings.

As a result, many anomaly detection methods treated even clean test images as abnormal because they differed from the learned normal training distribution. This increased false positives and reduced the separation between clean and artifact classes.

This is a major outcome because it shows that MRI artifact detection is not just an anomaly detection problem, but also a domain-shift problem. Any practical solution must therefore handle both artifact variation and acquisition-domain variation simultaneously.

11. Mild Artifact Detection Remains Largely Unsolved

The study also found that mild artifacts are considerably harder to detect than severe artifacts. Even the strongest performing method, PatchCore-kNN, achieved only modest performance for mild MR-ART cases. This is especially important from a clinical perspective because mild artifacts are often the most difficult and most relevant cases in actual radiological workflows.

The inability of current models to detect subtle degradations reliably suggests that clinically deployable artifact detection systems still require substantial improvement, particularly for borderline-quality scans.

6. Conclusion

This paper presented a systematic comparative evaluation of supervised, self-supervised, and unsupervised anomaly detection paradigms for MRI motion artifact detection, evaluated across two real-world datasets (MR-ART and KMAR), a simulated artifact set, and three training anatomy variants. Rather than confirming the anticipated hierarchy: supervised methods leading, label-free methods trailing, the results exposed a more nuanced and sobering picture: no method achieved strong discriminative performance on real MRI motion artifacts under the evaluated conditions, with most models recording AUROC values indistinguishable from chance on the primary test sets. The supervised ResNet-18 baseline failed, achieving test-set AUROC of 0.499–0.505 despite perfect validation performance (AUROC = 1.000 from epoch 1 on the brain variant). Post-hoc analysis of the training setup reveals this as a textbook case of shortcut learning: clean training images sourced from IXI and FastMRI share a different acquisition domain from artifact images sourced from MR-ART and KMAR, making dataset membership a trivially learnable proxy for the artifact label. This finding serves as a concrete methodological warning: supervised artifact detection benchmarks must employ acquisition-matched clean/artifact pairs, or reported performance will reflect domain classification rather than artifact sensitivity. Among the label-free methods, PatchCore with global kNN scoring was the sole method to produce a meaningful above-chance signal on real artifacts (AUROC = 0.673 on MR-ART, brain-trained), attributable to its frozen ImageNet backbone remaining uncontaminated by the source-domain confound that affected all gradient-trained encoders. Self-supervised methods — SimCLR, MAE, and DINO — converged successfully and showed healthy representations (with the exception of DINO on the narrow brain-only corpus, where representational collapse occurred and was remediated by combined anatomy training), but their anomaly scores were insufficiently discriminative for real motion artifacts under the 20-epoch training budget and existing scoring strategies. A striking dissociation emerged between synthetic and real artifact detection: reconstruction-based scorers achieved near-perfect detection of digitally imposed corruptions (MAE-Recon: 98.9%) while remaining near-chance on real k-space motion artifacts (AUROC = 0.459), demonstrating that synthetic benchmarks are not reliable proxies for clinical artifact detection performance.

Taken together, these results establish three concrete contributions. First, they provide the most thorough multi-paradigm comparison of anomaly detection approaches for MRI artifact detection to date under a rigorously controlled and matched experimental protocol. Second, they expose and characterize two failure modes, supervised shortcut learning and the synthetic-real artifact gap, that

have likely gone unreported in prior work due to insufficient evaluation rigor. Third, they demonstrate that representational collapse is a non-trivial risk in self-distillation applied to narrow medical imaging corpora, and that multi-anatomy training is a practical and effective mitigation requiring no architectural changes.

Future work should address the identified failure modes directly. On the supervised side, acquisition-matched clean/artifact pairs, whether from prospective paired scanning protocols or k-space-level motion simulation applied to clean images, are a prerequisite for meaningful benchmarking. On the label-free side, extending the training budget to 100–200 epochs for ViT-based methods (consistent with established practice in the self-supervised learning literature), incorporating k-space-aware artifact simulation into the anomaly scoring pipeline, and exploring anatomy-conditioned reference manifolds are promising directions. More broadly, the gap between 0.673 AUROC on severe brain motion artifacts and the near-chance performance on mild artifacts underscores that robust, clinically deployable MRI quality control remains an open problem — one whose solution will require closer integration between MRI physics, acquisition-domain awareness, and representation learning methodology.

7. References

1. Graves, M.J. and Mitchell, D.G. (2013), “Body MRI artifacts in clinical practice: A physicist's and radiologist's perspective,” *Journal of Magnetic Resonance Imaging*, vol. 38, pp. 269–287. <https://doi.org/10.1002/jmri.24288>
2. Katti, G., Ara, S., and Shireen, S. (2011), “Magnetic Resonance Imaging (MRI) - A review,” *International Journal of Dental Clinics*, vol. 3.
3. Kastrulin, S., Zakirov, J., Pezzotti, N., and Dylov, D. V. (2023), “Image Quality Assessment for Magnetic Resonance Imaging,” *IEEE Access*, vol. 11, pp. 14154–14168. <https://doi.org/10.1109/ACCESS.2023.3243466>
4. Lei, K., Syed, A. B., Zhu, X., Pauly, J. M., and Vasanaawala, S. S. (2022), “Artifact- and content-specific quality assessment for MRI with image rulers,” *Medical Image Analysis*, vol. 77, 102344. <https://doi.org/10.1016/j.media.2021.102344>
5. Jerald, A., Shanbhag, D., and Chatterjee, S. (2025), “Automated Motion Artifact Check for MRI (AutoMAC-MRI): An Interpretable Framework for Motion Artifact Detection and Severity Assessment,” *arXiv preprint arXiv:2512.15315*.
6. Litjens, G., Kooi, T., Bejnordi, B. E., Setio, A. A. A., Ciompi, F., Ghafoorian, M., van der Laak, J. A., van Ginneken, B., and Sánchez, C. I. (2017), “A survey on deep learning in medical image analysis,” *arXiv:1702.05747*. <https://doi.org/10.48550/arXiv.1702.05747>
7. Pizarro, R., Assemblal, H.-E., Jegathambal, S. K. B., Jubault, T., Antel, S., Arnold, D., and Shmuel, A. (2023), “Deep learning, data ramping, and uncertainty estimation for detecting artifacts in large, imbalanced databases of MRI images,” *Medical Image Analysis*, vol. 90, 102942. <https://doi.org/10.1016/j.media.2023.102942>
8. Zuo, L., Xue, Y., Dewey, B. E., Liu, Y., Prince, J. L., and Carass, A. (2023), “A latent space for unsupervised MR image quality control via artifact assessment,” *arXiv preprint arXiv:2302.00528*. <https://doi.org/10.48550/arXiv.2302.00528>
9. Belton, N., Tsighe Hagos, M., Lawlor, A., and Curran, K. M. (2024), “Towards a unified approach for unsupervised brain MRI motion artefact detection with few shot anomaly detection,” *Computerized Medical Imaging and Graphics*, vol. 115, 102391. <https://doi.org/10.1016/j.compmedimag.2024.102391>
10. Andre, J. B., Bresnahan, B. W., Mossa-Basha, M., Hoff, M. N., Smith, C. P., Anzai, Y., and Cohen, W. A. (2015), “Toward Quantifying the Prevalence, Severity, and Cost Associated

- with Patient Motion During Clinical MR Examinations,” *Journal of the American College of Radiology*, vol. 12, no. 7, pp. 689–695. <https://doi.org/10.1016/j.jacr.2015.03.007>
11. Nárai, Á., Hermann, P., Auer, T., Kemenczky, P., Szalma, J., Homolya, I., Somogyi, E., Vakli, P., Weiss, B., and Vidnyánszky, Z. (2022), “Movement-related artefacts (MR-ART) dataset,” *OpenNeuro*. <https://doi.org/10.18112/openneuro.ds004173.v1.0.2>
 12. Knoll, F. et al. (2018), “fastMRI: An Open Dataset and Benchmarks for Accelerated MRI,” arXiv:1811.08839.
 13. IXI Dataset. Available at: <https://brain-development.org/ixi-dataset/>
 14. He, K., Zhang, X., Ren, S., and Sun, J. (2016), “Deep residual learning for image recognition,” in *Proceedings of CVPR*, pp. 770–778.
 15. Küstner, S. et al. (2018), “Automated reference-free detection of motion artifacts in magnetic resonance images,” *Magnetic Resonance in Medicine*, vol. 80, no. 4, pp. 1537–1552.
 16. Tan, M. and Le, Q. (2019), “EfficientNet: Rethinking model scaling for convolutional neural networks,” in *Proceedings of ICML*, pp. 6105–6114.
 17. Kingma, D. P. and Welling, M. (2014), “Auto-encoding variational Bayes,” in *Proceedings of ICLR*.
 18. Baur, A. et al. (2018), “Deep autoencoding models for unsupervised anomaly segmentation in brain MR images,” in *MICCAI BrainLes Workshop*, pp. 161–169.
 19. Roth, K. et al. (2022), “Towards total recall in industrial anomaly detection,” in *Proceedings of CVPR*, pp. 14318–14328.
 20. Chen, T., Kornblith, S., Norouzi, M., and Hinton, G. (2020), “A simple framework for contrastive learning of visual representations,” in *Proceedings of ICML*, pp. 1597–1607.
 21. Caron, M. et al. (2021), “Emerging properties in self-supervised vision transformers,” in *Proceedings of ICCV*, pp. 9650–9660.
 22. He, K. et al. (2022), “Masked autoencoders are scalable vision learners,” in *Proceedings of CVPR*, pp. 16000–16009.
 23. Xi, Y., Wang, F., Shi, F., Zhou, Q., Li, R., Li, Y., and Wang, Y. (2025), “A Multi-view Open-access Dataset of Paired Knee MRI for Motion Artifact Removal,” *Scientific Data*, vol. 12. <https://doi.org/10.1038/s41597-025-05439-1>

24. Lippenszky, L., Yang, H., de Arcos, J., and Rusko, L. (2024), “Unsupervised Artifact Detection and Quantification via Contrastive Learning with Noise Reference,” in *Medical Imaging / Conference Paper*.
25. Lo, S.-C., Lou, S.-L., Lin, J.-S., Freedman, M. T., Chien, M. V., and Mun, S. K. (1995), “Artificial convolution neural network techniques and applications for lung nodule detection,” *IEEE Transactions on Medical Imaging*, vol. 14, pp. 711–718.
26. Prasoon, A., Petersen, K., Igel, C., Lauze, F., Dam, E., and Nielsen, M. (2013), “Deep feature learning for knee cartilage segmentation using a triplanar convolutional neural network,” in *MICCAI*, vol. 8150, pp. 246–253.
27. Herath, H. M. S. S., Herath, H. M. K. K. M. B., Madusanka, N., and Lee, B.-I. (2025), “A Systematic Review of Medical Image Quality Assessment,” *Journal of Imaging*, vol. 11, no. 4, p. 100. <https://doi.org/10.3390/jimaging11040100>
28. Mittal, A., Moorthy, A. K., and Bovik, A. C. (2011), “Blind/Referenceless Image Spatial Quality Evaluator,” in *ASILOMAR*, pp. 723–727. <https://doi.org/10.1109/ACSSC.2011.6190099>
29. Yang, P., Sturtz, J., and Qingge, L. (2023), “Progress in Blind Image Quality Assessment: A Brief Review,” *Mathematics*, vol. 11, no. 12, 2766. <https://doi.org/10.3390/math11122766>
30. Ma, J., Chen, Y., Chen, L., and Tang, Z. (2024), “Dual-attention pyramid transformer network for No-Reference Image Quality Assessment,” *Expert Systems with Applications*, vol. 257, 125008. <https://doi.org/10.1016/j.eswa.2024.125008>
31. Yu, S., Dai, G., Wang, Z. et al. (2018), “A consistency evaluation of signal-to-noise ratio in the quality assessment of human brain magnetic resonance images,” *BMC Medical Imaging*, vol. 18, 17. <https://doi.org/10.1186/s12880-018-0256-6>
32. Gao, R., Luo, G., Ding, R. et al. (2023), “A Lightweight Deep Learning Framework for Automatic MRI Data Sorting and Artifacts Detection,” *Journal of Medical Systems*, vol. 47, 124. <https://doi.org/10.1007/s10916-023-02017-z>
33. Esses, S. J., Lu, X., Zhao, T., Shanbhogue, K., Dane, B., Bruno, M., and Chandarana, H. (2018), “Automated image quality evaluation of T2-weighted liver MRI utilizing deep learning architecture,” *Journal of Magnetic Resonance Imaging*, vol. 47, pp. 723–728. <https://doi.org/10.1002/jmri.25779>

34. Vakli, P., Weiss, B., Szalma, J., Barsi, P., Gyuricza, I., Kemenczky, P., Somogyi, E., Nárai, Á., Gál, V., Hermann, P., and Vidnyánszky, Z. (2023), “Automatic brain MRI motion artifact detection based on end-to-end deep learning is similarly effective as traditional machine learning trained on image quality metrics,” *Medical Image Analysis*, vol. 88, 102850. <https://doi.org/10.1016/j.media.2023.102850>
35. Wang, Y. and Shi, F. (2025), “KMAR-50K,” *Mendeley Data*, V6. <https://doi.org/10.17632/xw7mrg7ntg.6>
36. Ali, M. A., Dornaika, F., Arganda-Carreras, I., Chmouri, R., and Shayeh, H. (2024), “Enhancing MRI brain tumor classification: A comprehensive approach integrating real-life scenario simulation and augmentation techniques,” *Physica Medica*, vol. 127. <https://doi.org/10.1016/j.ejmp.2024.1048>
37. He, K., Zhang, X., Ren, S., and Sun, J. (2015), “Deep residual learning for image recognition,” arXiv:1512.03385.
38. Schaumberg, A. J., Rubin, M. A., and Fuchs, T. J. (2016), “H&E-stained whole slide deep learning predicts SPOP mutation state in prostate cancer,” *bioRxiv*:064279.
39. Chen, H., Dou, Q., Yu, L., and Heng, P.-A. (2016), “VoxResNet: Deep voxelwise residual networks for volumetric brain segmentation,” arXiv:1608.05895.